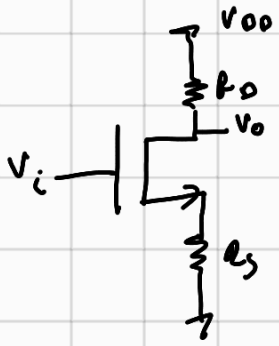
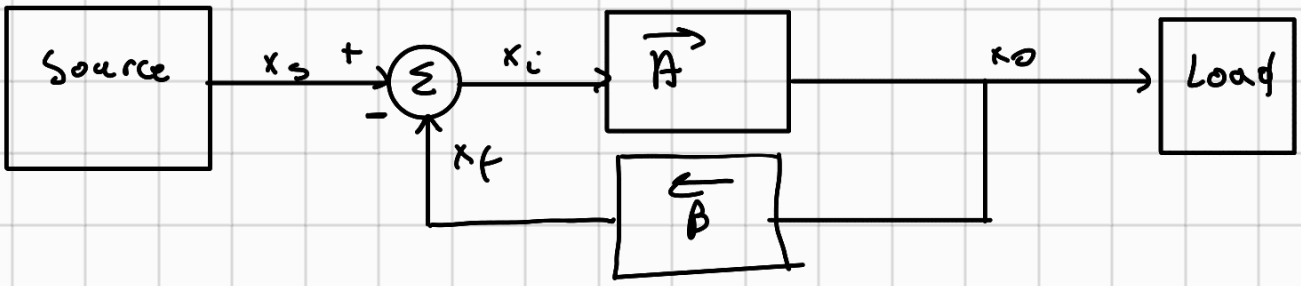
Feedback



Common source amplifier

$$\frac{v_o}{v_i} \simeq \frac{R_D}{R_S}$$

The general feedback Structure



A = open loop gain

β = feedback factor

x_i = Input signal

x_s = signal source

x_o = output signal

x_f = sampled output by fbn ↓ feedback network

$\left(\frac{x_o}{x_s}\right)_{CL} = ?$

$x_o = A x_i$

$x_f = \beta \cdot x_o$

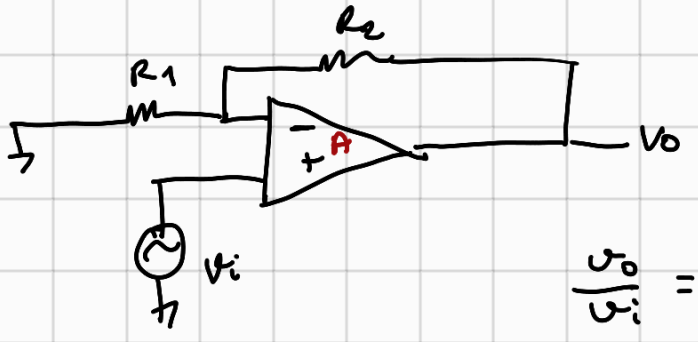
$x_i = x_s - x_f$

$x_i = x_s - \beta x_o$

$X_i = x_s - A\beta \cdot x_i$

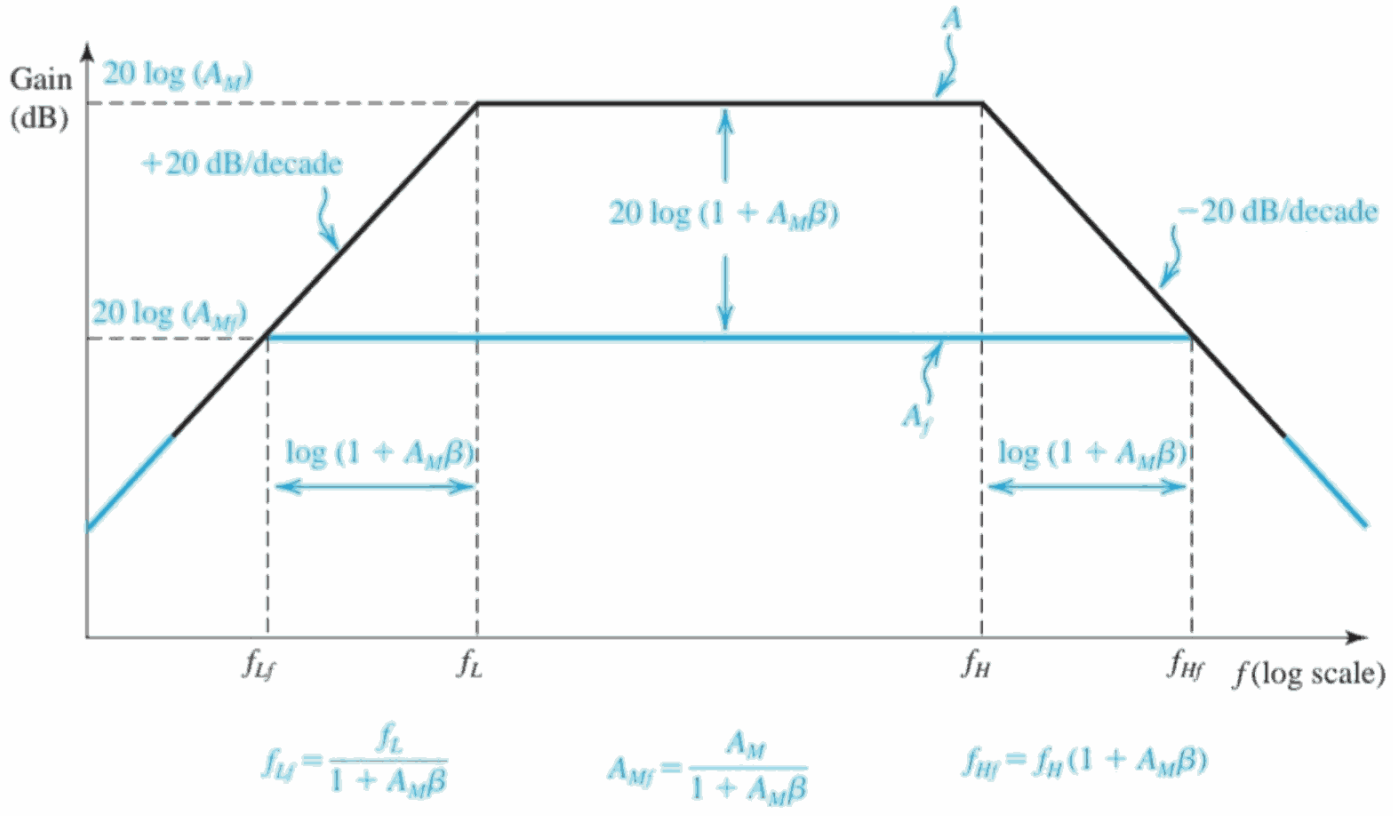
$x_s = x_i (1 + A\beta)$

$$\left(\frac{x_o}{x_s}\right) = \frac{A \cancel{x_i}}{\cancel{x_i}(1+A\beta)} = \frac{A}{1+A\beta}$$



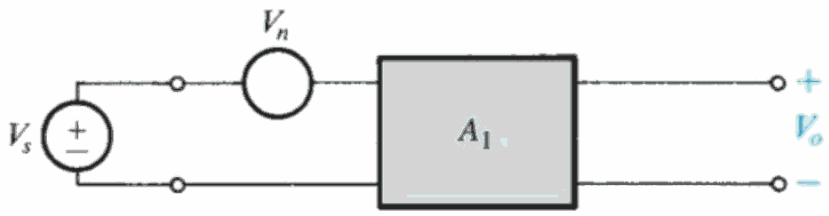
$$\frac{v_o}{v_i} = \left(1 + \frac{R_2}{R_1}\right) = 1/\beta$$

$\beta = \left(1 + R_2/R_1\right)^{-1}$

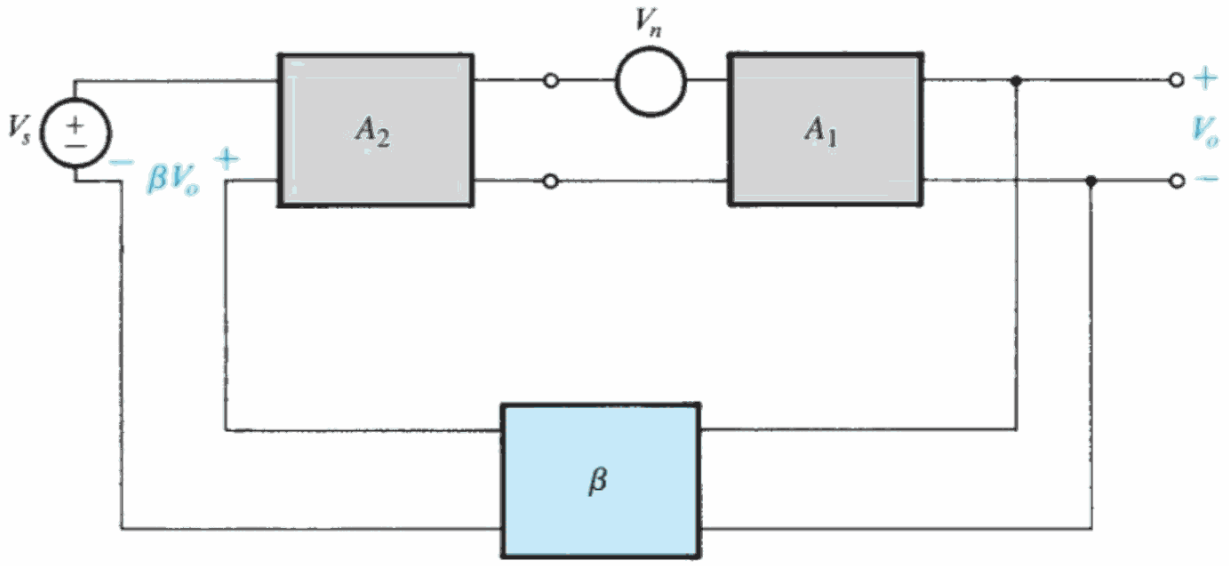


$$f_{Lf} = \frac{f_L}{1 + A_M\beta} \qquad A_{Mf} = \frac{A_M}{1 + A_M\beta} \qquad f_{Hf} = f_H(1 + A_M\beta)$$

Figure 10.3 Application of negative feedback reduces the midband gain, increases f_H, and reduces f_L, all by the same factor, $(1+A_M\beta)$, which is equal to the amount of feedback.



(a)



(b)

Voltage amplifiers

Shunt - Series
(O) (I)

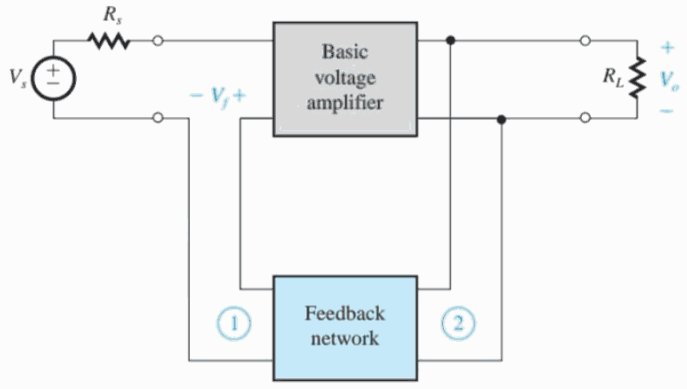
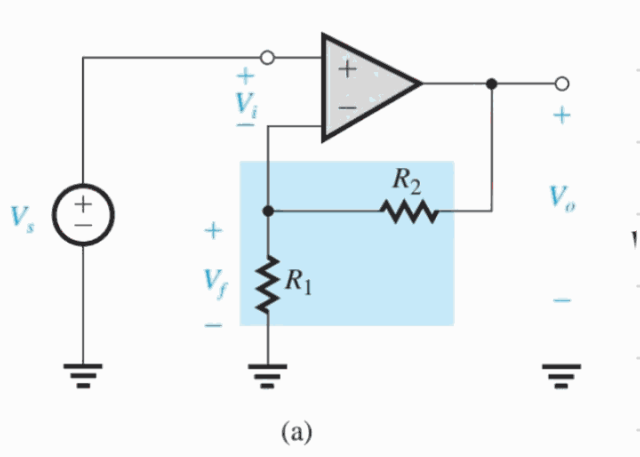
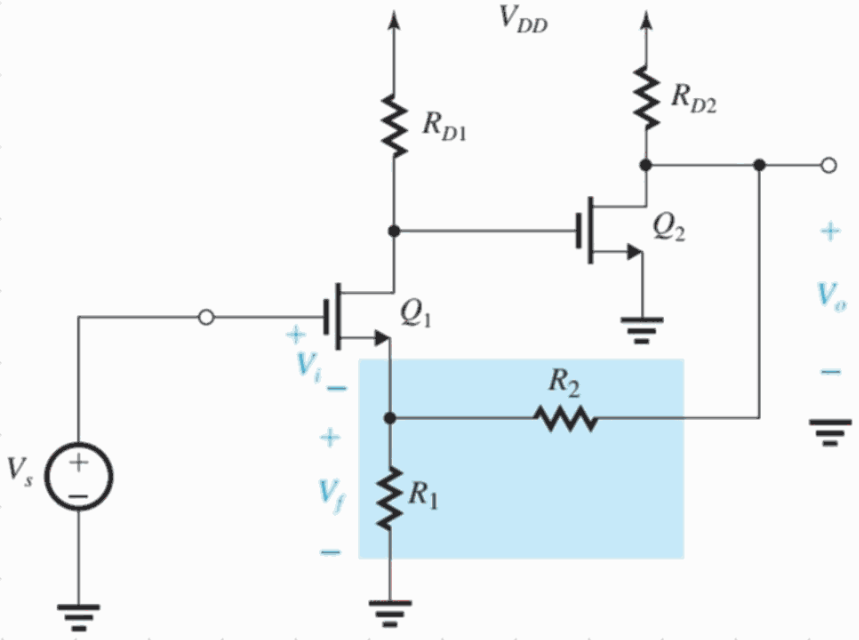
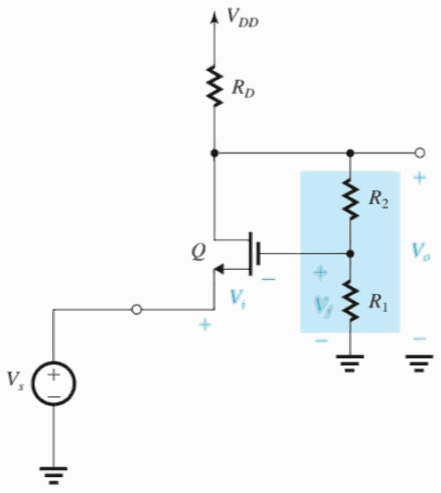


Figure 10.6 Block diagram of a feedback voltage amplifier. Here the appropriate feedback topology is series–shunt.

This effects lower output resistance





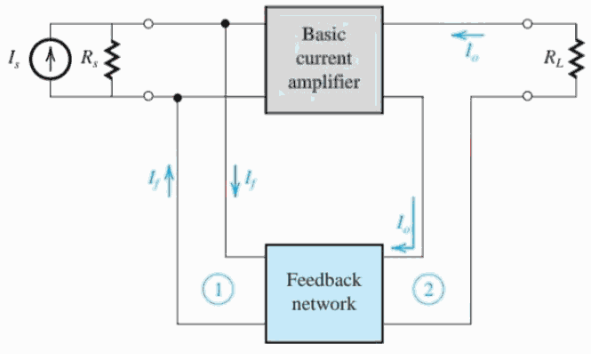


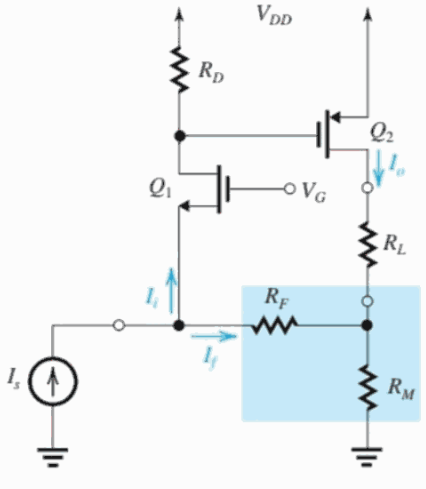
(c)

Figure 10.7 *continued*

Current Amplifier

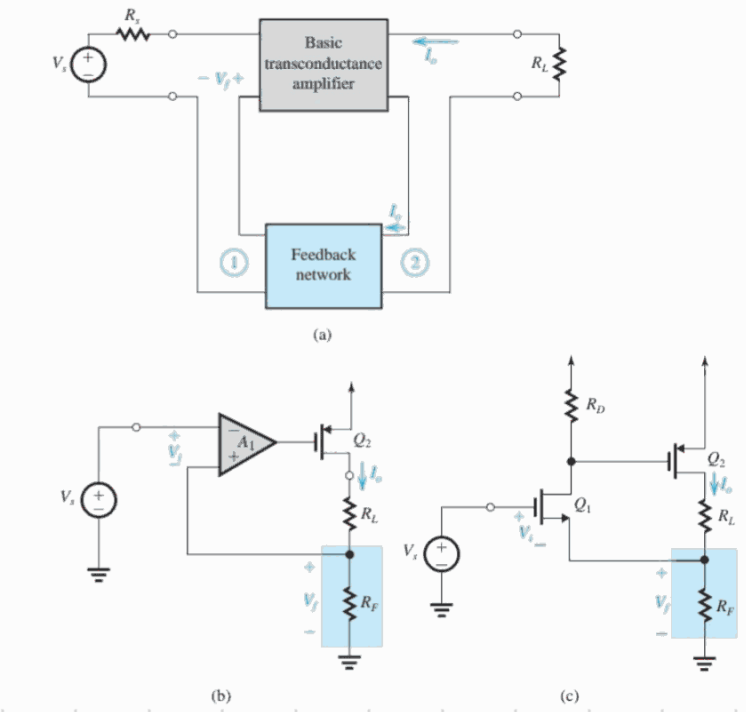
Series - Shunt
(o) (i)



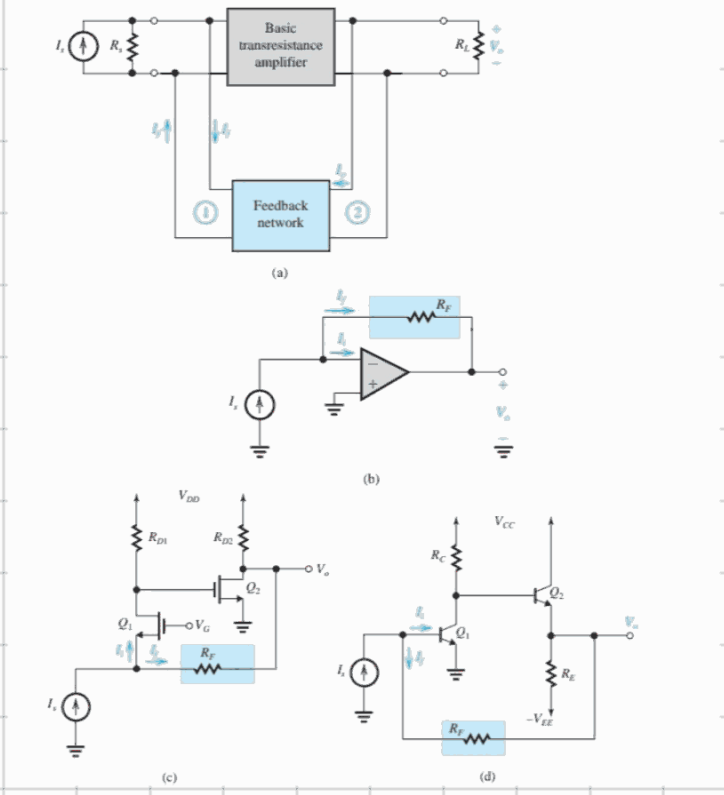


Transconductance amplifiers

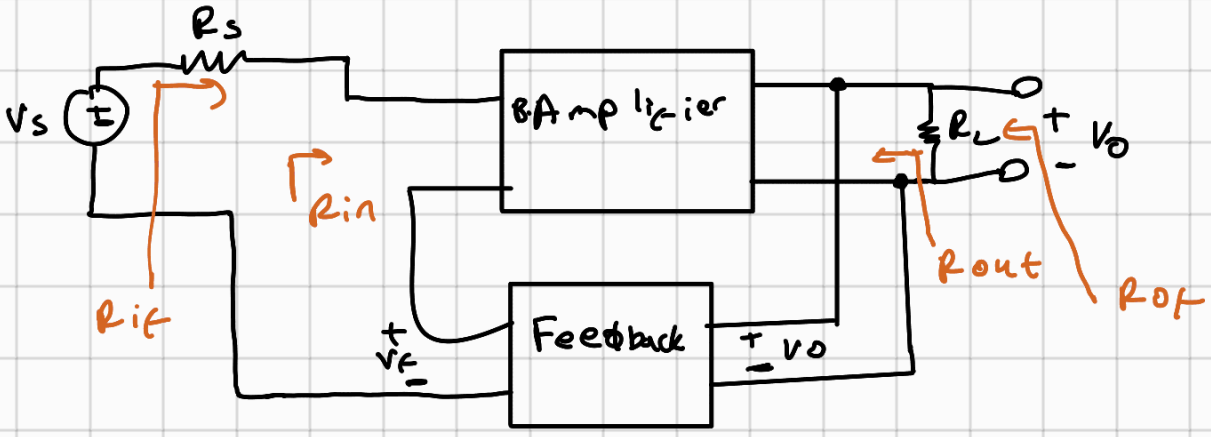
Series Series



Transresistance Amplifiers



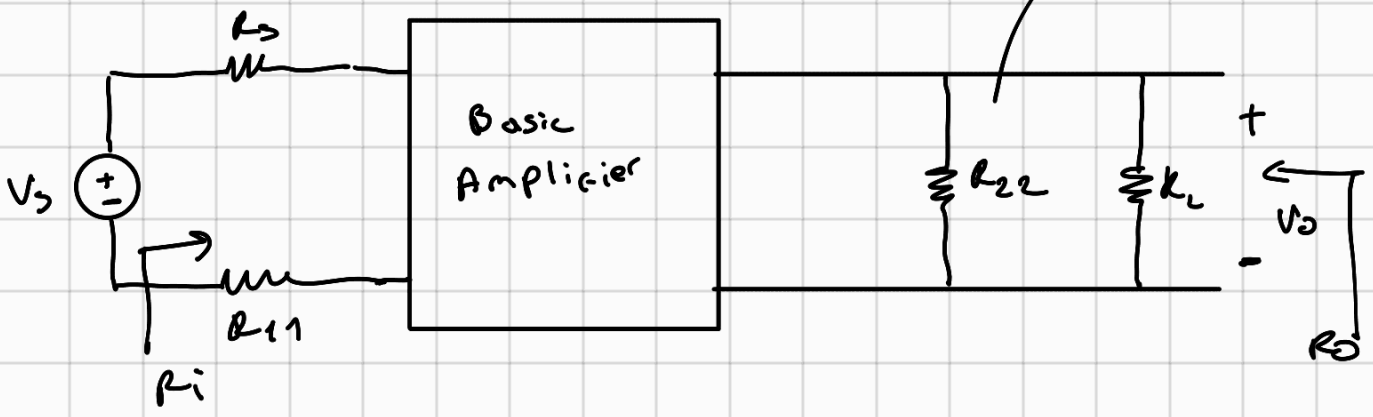
1- Voltage Amplifiers (Shunt - Series)
(O) (I)

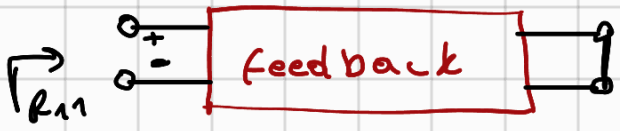


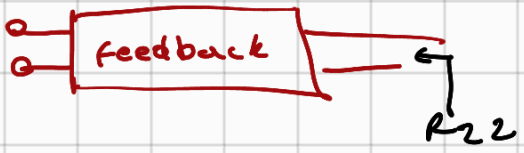
Two port network equivalent

feed forword gain

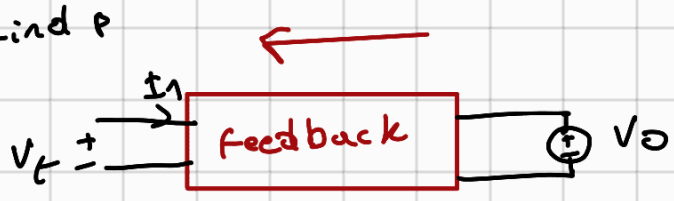
$A = \frac{v_o}{v_s}$





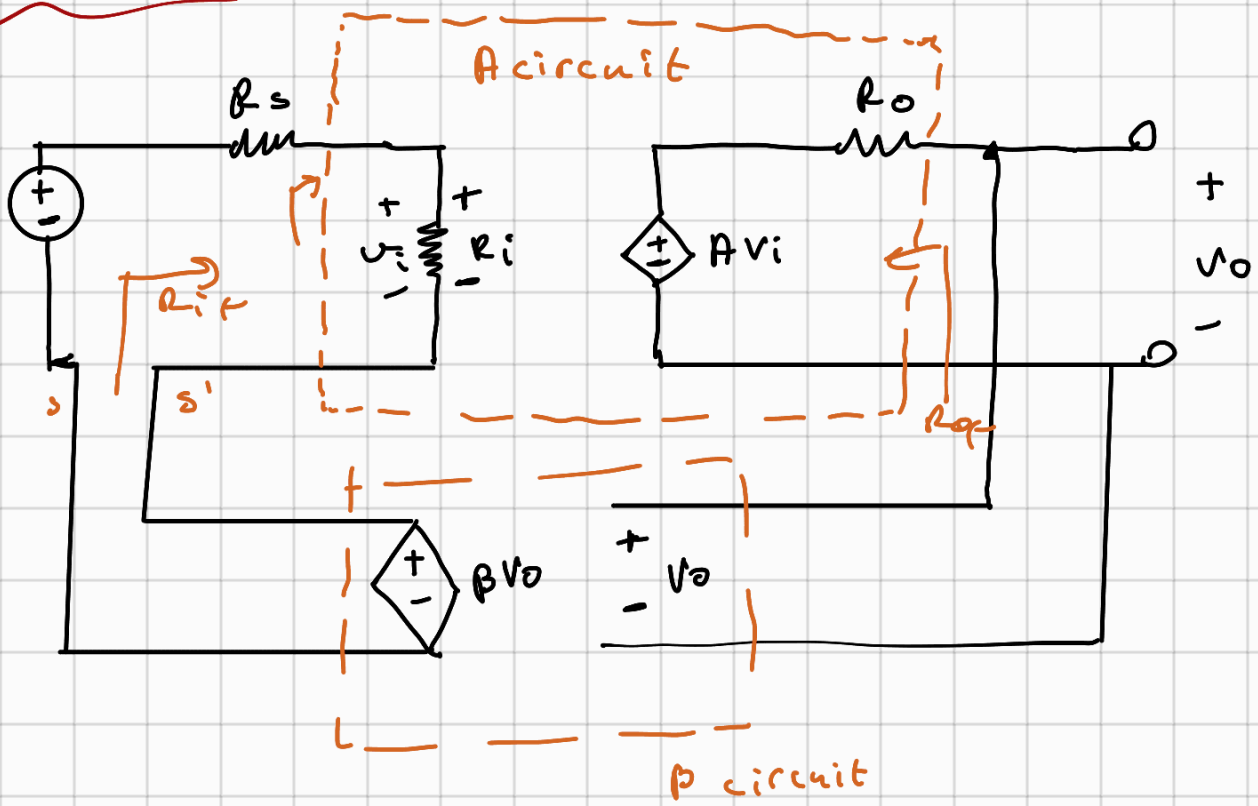


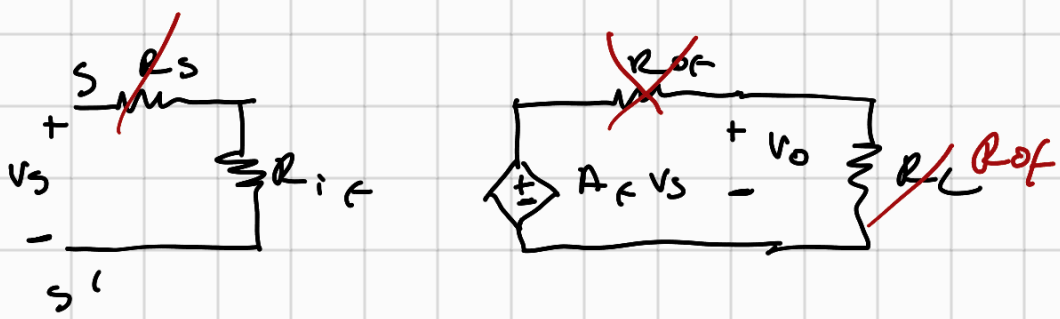
To find β



$$\beta = \frac{V_f}{V_o}\Big|_{I_1=0}$$

Complete model





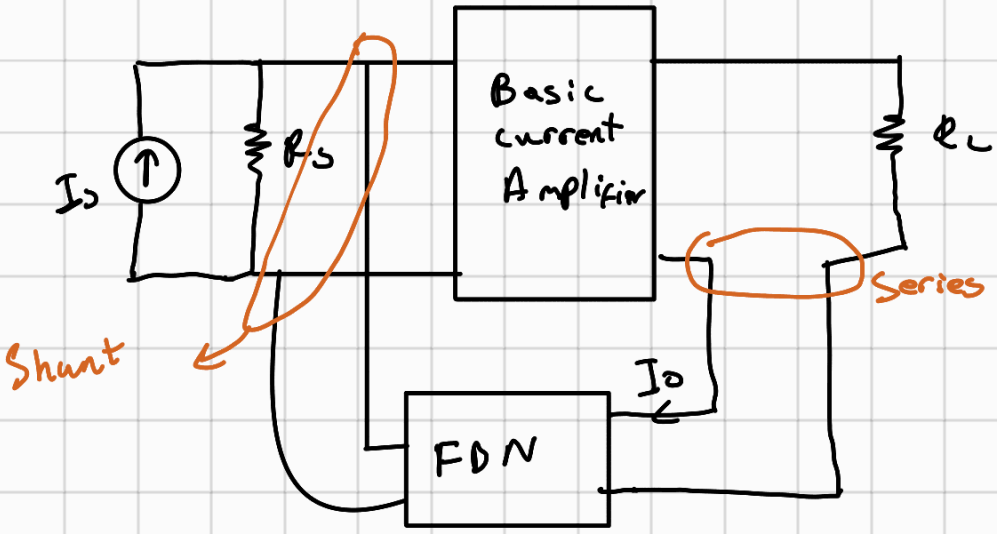
$$A_f = \frac{A}{1+A\beta}$$

$$R_{if} = (1+A\beta)\,R_i$$

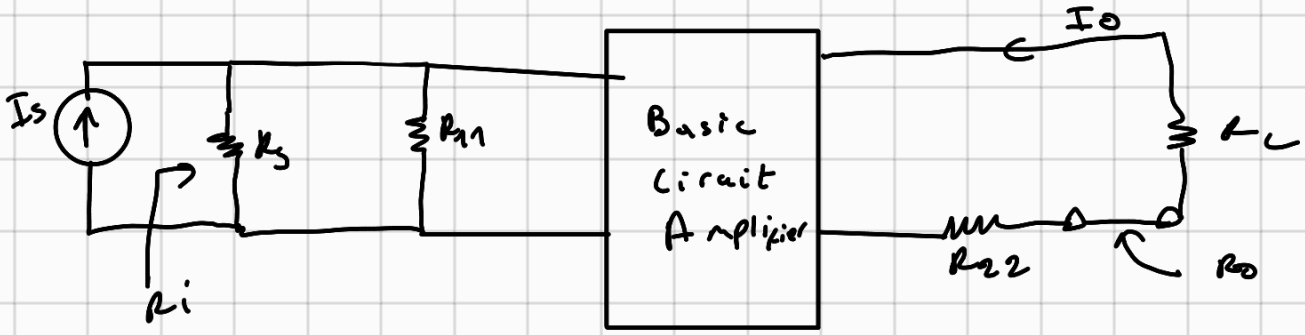
$$R_{of} = \frac{R_o}{1+A\beta}$$

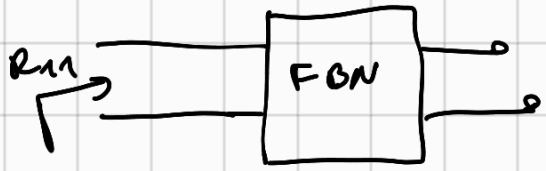
2- Current Amplifiers (Series - Shunt)

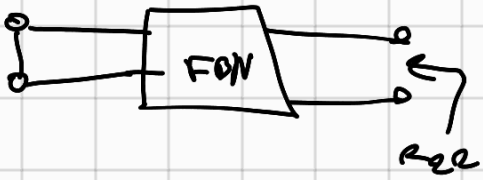
(o) (I)

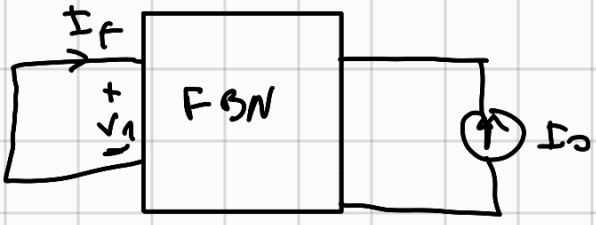


$$A = \frac{I_o}{I_i}$$

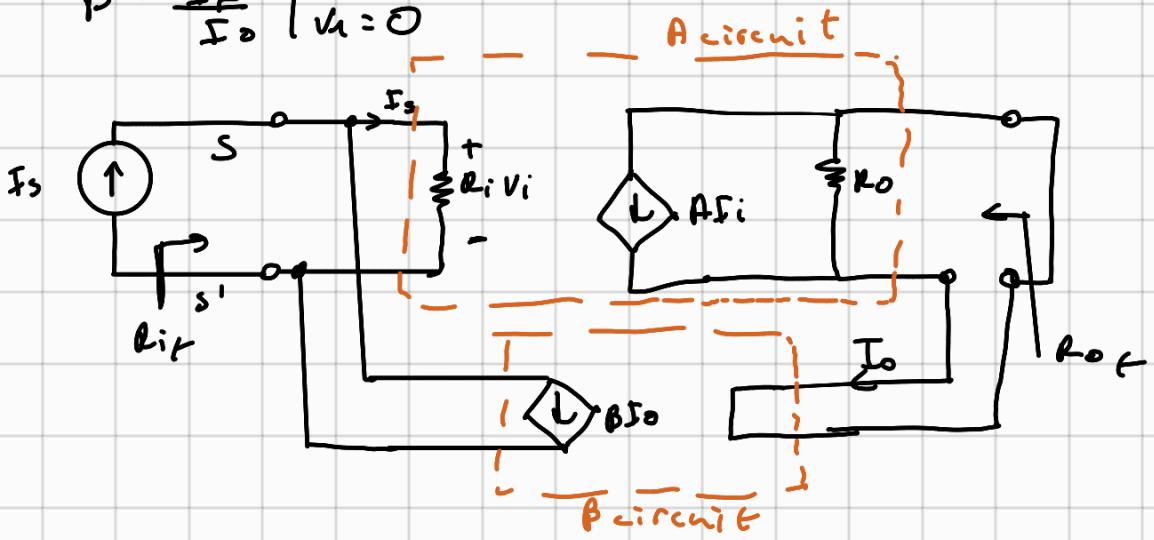


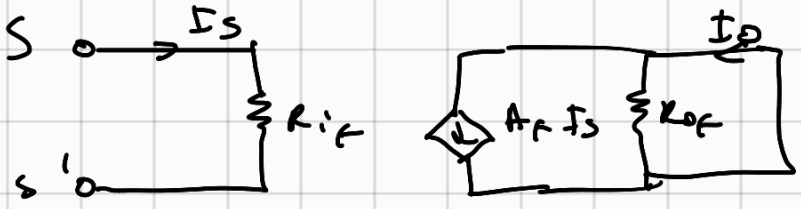






$$\beta = \frac{I_f}{I_o}\Big|_{V_1 = 0}$$





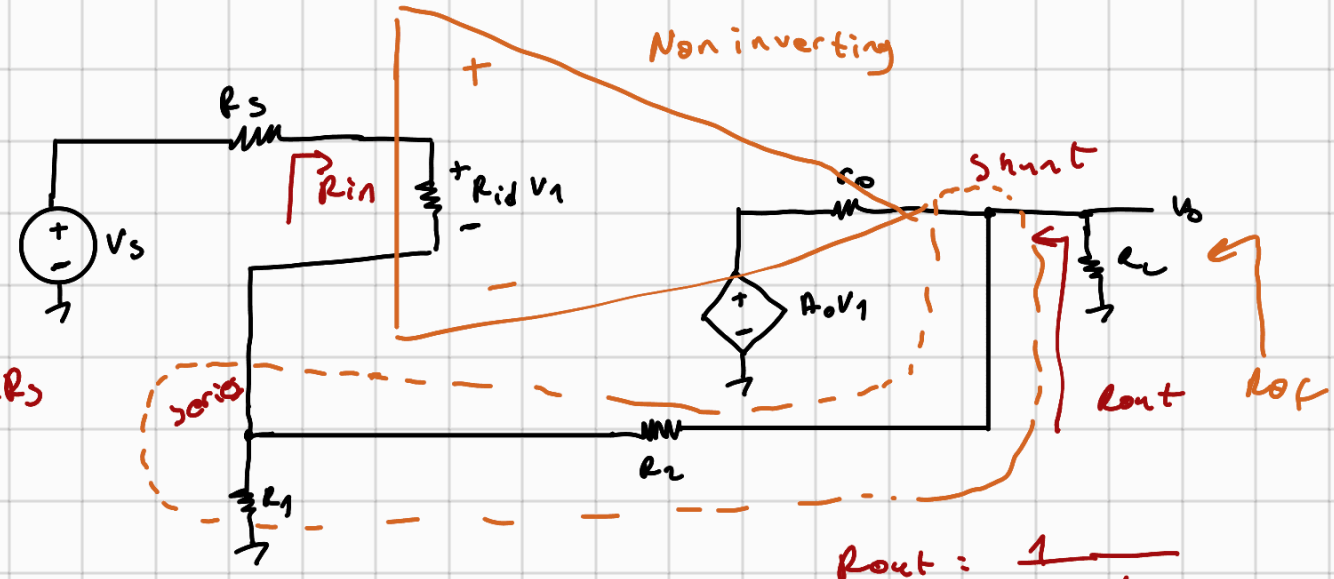
$$A_f = \frac{A}{1+A\beta}$$

$$R_{if} = R_i / 1+A\beta$$

$$R_{of} = R_o(1+A\beta)$$

$$1+\frac{R_2}{R_1} = 1001$$

Soru



$R_{in} = R_{if} - R_s$

$$R_{out} = \frac{1}{1/R_{of} - 1/R_L}$$

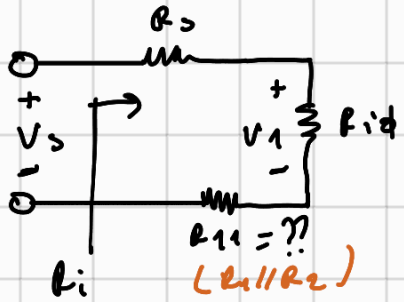
Find expressions for

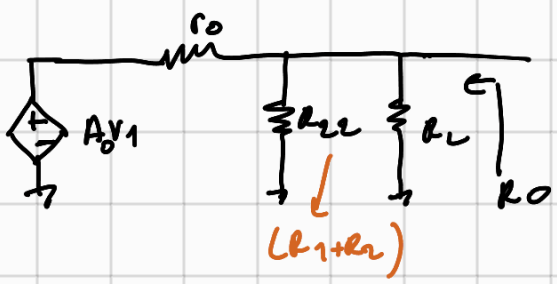
$A, \beta, A_f, R_i, R_{if}, R_{of}$, and R_o

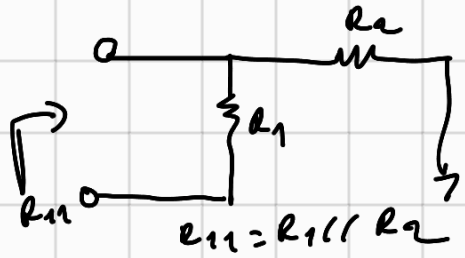
($A_o = 10^4$, $R_{id} = 100\,k\Omega$, $r_o = 1\,k\Omega$, $R_L = 2\,k\Omega$

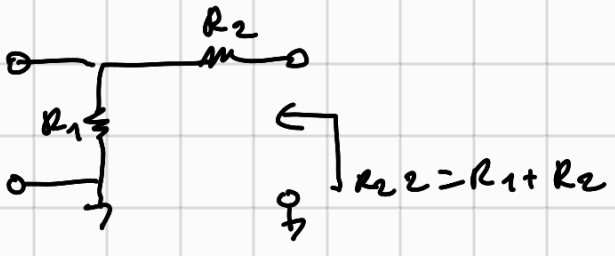
$R_1 = 1\,k\Omega$, $R_2 = 1\,Meg\Omega$ and $R_s = 10\,k\Omega$)

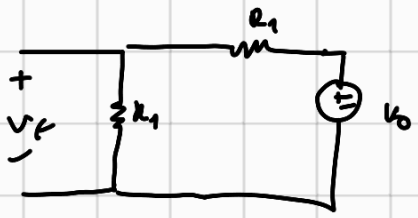
Feedback-type : Shunt-Series











$$\frac{V_f}{V_O} = \frac{R_1}{R_1+R_2} = \beta$$

$$R_i = R_s + R_{id} + R_{11}$$

$$R_o = r_o \,//\, R_{22} \,//\, R_L$$

$$A = \underbrace{\frac{r_{id}}{R_i}}_{\frac{V_o}{V_s} = \frac{V_1}{V_s}} \cdot \underbrace{\frac{R_{22} // R_L}{r_o + R_{22} // R_L}}_{\frac{V_o}{V_1}\ \text{katkıları}} \cdot A_o$$

$$A_f = \frac{A}{1 + A\beta}$$

$$A = \frac{100k\Omega}{111k\Omega} \cdot \frac{2k\Omega}{3k\Omega} \cdot 10^4 \simeq 6000\ [V/V]$$

$$\beta = \frac{R_1}{R_1 + R_2} = 10^{-3}\ [V/V]$$

$R_i = 111k$

$R_o = 0,60k$

$$1 + A\beta = 1 + 6000 \cdot 10^{-3} = 7$$

$$A_f = \frac{6000}{7} = 857\ [V/V]$$

$$R_{if} = (1 + A\beta) \overset{111k}{R_i} = 777k\Omega$$

$$R_{of} = \frac{R_o}{(1+A\beta)} \simeq 93\Omega$$